

Mid Sem

Course: Principles of Economics

Course Code: ECO 201

Total points: 30

Weightage towards final grade: 30%

Date: 27/02/2026

Time: 3:00 PM - 4:10 PM

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Section I

Total: _____

21. Back Page

22. Back Page

23. _____

24. _____

Section II=

Total: _____

Section I Total: _____ **+ Section II Total:** _____

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A 2D coordinate system is shown with a horizontal axis labeled x_2 and a vertical axis. The origin is at the bottom-left corner.

A blank Cartesian coordinate system with a horizontal x-axis and a vertical y-axis. The x-axis is labeled with the variable x at its right end, and the y-axis is labeled with the variable y at its top end. The two axes meet at an origin in the bottom-left corner, forming an L-shape. There are no tick marks or grid lines on the axes.[illegible]

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The context/clarifications page will only count towards questions that are phrased ambiguously. It is not for questions which you are unable to interpret properly. For example, in Quiz 2, only the first question would count for this.

Section I (1 mark per question, no negative marking, 15% of grade, CO1 & CO2)

1. Which of the following is not true

- (A) Absolute advantage can exist without a relative advantage
- (B) Relative advantage can exist without absolute advantage
- (C) Both (A) and (B) (D) NOTA

Answer. (B) is not true. If you do not have an absolute advantage in the production of at least one good, there is no question of relative advantage as opportunity costs are identical. However (A) can happen if, for example, linear production possibility frontiers are parallel. However, there is some confusion in the interpretation of option (B), i.e., whether it is an absolute advantage of a system or an individual, thus, I am considering (C) as well.

2. In the context of GDP, when prices increase (other things remaining constant)

- (A) Real GDP decreases (B) Nominal GDP decreases
- (C) Nominal GDP increases (D) Both (A) and (C)

Answer. (C). Real GDP does not change. It will remain the same once corrected for price change.

3. A price ceiling p'

(A) is below the equilibrium price (B) is above the equilibrium price

(C) Causes a quantity shortage (D) Both (A) and (C)

(E) Need more information

Answer. (E) To answer this question we need to know if the ceiling is binding or not.

4. Deadweight loss due to tax is 0 when demand is _____

Answer. Perfectly inelastic (not quantity change)

5. Consider a perfectly inelastic supply curve and a typical downward sloping linear demand curve. Suppose the demand curve shifts outward (other things remaining same). What happens to CS

(A) Increases (B) Decreases (C) Remains same.

(D) Depends on slope of demand

Answer. (C) Remains same. The area of the triangle determining CS does not change

6. The production set takes costs of inputs as given/fixed

(A) True (B) False (C) Production set does not, but production function does.

Answer. (B) False. The production function is independent of prices. Profits/Revenues are a different matter

7. Coase Theorem suggests that private market can solve the problem of externalities if they can bargain over _____ at no cost

(A) Equilibrium price (B) Allocation (C) Consumption (D) NOTA

Answer. (B) Allocation. This is by definition

8. Goods that are free to use but impose cost externalities are known as

(A) Public Goods (B) Private goods (C) Common Pool resources

(D) Club goods

Answer. (C) Common Pool Resources. By definition

9. As the Stock S of a common pool resource becomes very large (goes to infinity), a common pool resource becomes a _____

Answer. A Public good. We saw this in the example in class. There are no cost externalities if the resource is very large.

10. Inventory is considered as

(A) Consumption (B) Investment (C) It is added separately

(D) It is not counted in the GDP as it will be counted in the next year/quarter etc.

Answer. (B) Investment

11. Which of these utility functions violates some of our basic assumptions (both x and y are positive)

(A) $U(x, y) = x + y$ (B) $U(x, y) = y \ln(x)$ (C) $U(x, y) = \frac{x}{1+y}$ (D) NOTA

Answer. (C) An increase in a good y is decreasing utility. This violates our basic assumption.

12. in the context of controlling negative externalities carbon credits are an example of

(A) Trading debt (B) Command-and-control (C) Trading quotas

(D) Taxes

Answer. (C) Trading quotas

13. In models of free-riding (generalized to exclude extreme cases), putting in effort does not add to utility at all. True or False?

Answer. This is false. It may not be optimal to put a lot of effort, but effort does increase utility.

14. Cross-price elasticity of demand is generally (without exceptional cases)

(A) Positive (B) Negative (C) 0 (D) Can be (A), (B) or (C)

(E) Can be (A) or (B) but not (C)

Answer. (D) Depends on whether goods are substitutes, complements or neither.

15. A unit elastic demand curve is

(A) Linear (B) Non-linear (C) Linear and flat (D) Linear and vertical.

Answer. Non-linear. You can check this in your text book.

16. Produced means of production are known as _____

Answer. Capital

17. In principle, for 2 goods x_1 and x_2 , what does a smooth indifference curve for given utility $\bar{k}$ represent?

(A) Sustainability (B) Affordability (C) Optimality (D) Price Elasticity

Answer. (G) NOTA (Options were updated in class). The answer is substitutability, which I mistyped in option (A).

18. Barring the special case when demand and supply curves are either perfectly elastic or inelastic, if we follow the standard assumptions on the slope of demand and supply, What is the minimum number of times the curves/lines will cross each other?

Answer. Zero. Since they are only in the positive quadrant, one can be above the other.

19. Which of these is true?

- (A) Increase in budget constraint shifts all IC to the right
(B) Increase in budget constraint shifts all IC to the left
(C) An IC intersects the budget constraint either once or twice. (D) NOTA

Answer. (D) NOTA. ICs do not shift because of budget constraints. And IC may not intersect a budget constraint at all.

20. Consider the budget set involving two goods x_1 and x_2 with corresponding prices p_1 and p_2 . We can calculate the area of the budget set on the graph. Suppose the area of three distinct budget sets is the same. This is possible when

- (A) The budget constraint is different. (B) Utility functions vary
(C) Not possible.

Answer. (C) Not possible. For a given p_1 and p_2 it is only possible to have different budget constraints if the budget changes. This will change the area as well.

Section II (2.5 marks per question, no negative marking, 10% of grade, CO1)

21. Show graphically why 2 indifference curves cannot intersect with proper labeling. You have to show why it would not make sense. Draw on back page.

Answer. You can show this in a graph where since ICs represent constant utility and higher ICs always have a higher utility, an intersection would contradict this. You can explain it easily as well.

22. Suppose there is only one input x and the output is y . Draw the isoquant (back page)

Answer. Isoquants are between the inputs. Any input x would also increase y , so a parallel line to the x axis would not work. However, if you have shown the isoquant as a dot, I can consider it.

23. Consider 4 possible allocations between citizen 1 and 2. A: (1,1), B: (2,0), C: (0,10) and D: (0,0). Which one of these is/are Pareto Efficient? Fill in the blank on your answer sheet. If the Answer is A and B, write A, B.

Answer. A, B and C are all Pareto Efficient as you cannot make anyone better without making someone worse off. Only (D) is not Pareto Efficient

24. Suppose there are two types of goods, high (H) and low (L) quality. A buyer is willing to pay up to b_i for a good where $i \in \{L, H\}$. Similarly, the seller is willing to sell the good at a minimum price of s_i . Suppose $s_L + \epsilon \geq b_L$ and $s_H + \epsilon < b_H$. $s_i, b_i, \epsilon > 0$ (ϵ is small).

- (A) Both goods should sell (B) No goods should sell
(C) Only low type should sell (D) Only high type should sell.
(E) Can't say for sure.

Answer. (E) Can't say for sure. Depends on whether not $s_L > b_L$ or not.

XXXXXXXXXXXXXXXXX END OF PAPER XXXXXXXXXXXXXXXXXXXX

Rough Work
